

Input Frame

Semantic Segmentation

Differentiable Geometric Registration

Template Projection

SAM 3

Court Prediction

Weighted Differentiable DLT



$$I \in \mathbb{R}^{H \times W \times 3}$$

Backbone
ViT (frozen)

FPN neck
fine-tuned

$3 \times 3 \times 2 \cdot 1 \times 1$

Conv

Visibility Head

GAP

Linear

$$\pi_k \in (0, 1)$$

Landmark Head

$$\hat{u}_k = \text{soft-argmax} \sum_p p H_k(p)$$



Court Segmentation (aux.)

$$\hat{u}_k$$

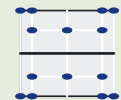
$$\pi_k$$



solve

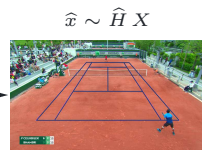
$\hat{H}$

Metric Court Template $\mathcal{T}$



$$Q = \{q_k\}_{k=1}^{14}$$

$$q_k \leftrightarrow \hat{u}_k$$



Registered Court
projected court lines

$$\hat{x} \sim \hat{H} X$$

Gradients Update